

Mayank Kumar

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EDUCATION

- **Indian Institute of Information Technology, Tiruchirappalli** Tamil Nadu, India
Bachelor of Technology in Electronics & Communication Engineering Aug 2023 – May 2027

PROJECTS

- **AI Workflow Automation Platform** GitHub | Live Demo
MERN, Redis, BullMQ, Socket.IO, Groq API
 - Engineered a background job execution engine (Node.js, Redis, BullMQ) with a topological-sort scheduler and Groq LLM API integration to process multi-step AI workflows asynchronously, reducing API response time by 95% (from a blocking 10+ second request to a sub-200ms queued response) via independent worker processes with real-time status updates through Socket.IO.
 - Built a full-stack visual workflow automation platform (React Flow drag-and-drop builder) with JWT-based auth (access/refresh token rotation) and a Node.js/Express/MongoDB backend.
 - Implemented 5 third-party integrations (GitHub, Slack, Gmail, Google Drive, Discord) with OAuth2 flows, encrypted token storage, and automatic token refresh.
 - Added real-time execution monitoring via Socket.IO (bridging a separate worker process through BullMQ's event queue) and public webhook triggers for external system integration.
 - Deployed on Render (API + worker) and Vercel (frontend) with production-grade security (encrypted credentials, CORS, secure cross-domain cookies).
- **DPI Engine – Deep Packet Inspection System** GitHub
C++17, Multithreading, CMake, Network Protocol Parsing, TLS/HTTP/DNS Inspection, PCAP, Python
 - Built a C++ Deep Packet Inspection engine that parses raw PCAP captures (Ethernet/IP/TCP/UDP) and extracts TLS SNI, HTTP Host headers, and DNS queries to classify traffic into 20+ real-world apps (YouTube, Netflix, WhatsApp, etc.) without decrypting HTTPS, achieving 100% flow identification accuracy in test captures via a multi-threaded (4-way parallel) load-balanced inspection pipeline.
 - Implemented flow-based IP/app/domain blocking using a five-tuple connection tracker to maintain per-flow state across packets.
 - Architected a multi-threaded pipeline (Reader → Load Balancer → Fast Path workers) with consistent hashing and thread-safe queues to parallelize inspection across configurable thread pools.
 - Achieved zero-dependency, cross-platform support (Linux/macOS/Windows) via a custom byte-order abstraction layer.

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, JavaScript
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Computer Networks, Operating Systems, Database Management Systems (DBMS)
- **Frontend:** HTML, CSS, React.js
- **Backend:** Node.js, Express.js, REST APIs, Flask
- **Databases:** MySQL, MongoDB, SQL
- **Tools & Platforms:** Git, GitHub, Docker, Postman, AWS

ACHIEVEMENTS & CODING PROFILES

- **Codeforces:** Rated Pupil, actively solving competitive programming problems.
- **CodeChef:** 3-Star rated, with a rating of 1602.
- **Decode X Hackathon:** Finalist, March 2026.
- **DSA Practice:** Solved 900+ Data Structures & Algorithms problems across platforms.